INCH-POound

MIL-C-24757(SH)
9 November 1990

MILITARY SPECIFICATION

CLOTH, FIBERGLASS, COATED

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of inherently fire-safe, shrink-safe, woven cloths.

1.2 Classification. Cloth covered by this specification shall be as specified (see 6.2):

100 percent fiberglass base material that is coated on both sides in accordance with paragraph 3.5

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4210
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STANDARDS

FEDERAL

FED-STD-4	Glossary of Fabric Imperfections
FED-STD-191	Textile Test Methods

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadeladelphia, PA 19111-5094)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

FEDERAL TRADE COMMISSION PUBLICATIONS

Textile Fiber Products Identification Act

(Copies may be obtained from the Federal Trade Commission, 6 Pennsylvania Avenue, Washington, DC 20580.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC Color Transference Chart

(Copies may be obtained from the American Association of Textile Chemists and Colorists, PO Box 12215, Research Triangle Park, NC 27709.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 578 Standard Specification for Glass Fiber Strands

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents may also be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4), in accordance with 4.3.

3.2 Material.

3.2.1 Fibers. Fiberglass fiber: The fiber shall be “C” filament type fiberglass (see 4.1.1 and 4.5).

3.2.2 Yarns. The warp and filling yams shall be singles (see 4.5).

3.2.3 Fiber content. The fiber content shall be as specified in 3.2.1 when tested in accordance with 4.5 (see 4.1.1).

33 Weave. The weave shall be of a type which does not promote potential fiber breakdown resulting in tensile strength loss after extended exposure to folded/creased storage. Plain and taffeta. weaves are not recommended.

3.4 Color. The color of the finished cloth shall approximate the standard sample that has been approved by the contracting officer. The finished cloth shall be gold in color (see 6.21).

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3.4.1 Colorfastness. The dyed cloth shall show fastness to laundering (after 3 cycles), mocking, and light (20 hours) equal to or better than the standard sample. When no standard sample has been established or designated as applicable to colorfastness, the dyed and finished cloth shall show “good” fastness to laundering (after 3 cycles), light (after 20 hours exposure) and shall show a Munsell Value for crocking, in accordance with AATCC Color transference chart, not lower than 8.5 (see 4.5).

3.5 Finish. The coating compound shall be an approved, acrylic or a urethane-acrylic aqueous polymeric system which is corrective with water repellent silicons and flame retardant elements. The compound shall be compatible with the base cloth to ensure its proper adherence to the yams of the base cloth. The coating shall be applied to both the face and back of the base cloth.

3.6 Shrinkage The finished cloth shall not shrink nor elongate more than 2 percent in the warp and falling directions when tested in accordance with 4.5.

3.7 Width. The width inclusive of salvage shall be as specified by the contracting activity with a plus or minus toerance of 2 percent (see 6.2).

3.8 Designs and patterns. The designs and patterns, dyed color or natural, shall be as specified (see 6.2).

3.9 Length and put-up. Unless otherwise specified (see 6.2) cloth shall be in one continuous length of 50 plus or minus 2-1/2 yards, put-up in rolls.

3.10 Physical requirements. The finished cloth shall conform to the requirements specified in table I when tested in accordance with table II.

3.11 Marking. The length of each roll shall be legibly marked with indelible ink on a ticket securely attached to the roll.

3.12 Workmanship. The finished cloth shall conform to the quality established herein. The demerit points per 100 square yards, when calculated as specified herein, shall not exceed the established maximum point value.

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TABLE I. *Physical requirements.*

Characteristics	Requirements
Weight ounce/square yard (minimum)	best cloth (minimum) 6.6
(gram/square meter)	finished cloth (maximum) 8.6
Yarns per inch (centimeter (cm)) (minimum)	
warp	(minimum) 38
Filling	(minimum) 38
Breaking strength pounds (newtons) (minimum)	
warp	(minimum) 250
Filling	(minimum) 250
Tearing strength pounds (newtons) (minimum)	
Warp	(minimum) 7.86 (35)
Falling	(minimum) 7.86 (35)
Flame resistance (warp and filling)	
Flaming time, seconds (maximum)	(maximum) 20
Glow time, seconds (maximum)	(maximum) 3.0
Char length, inches, cm (maximum) average	(maximum) 1.0
Hydrostatic sign (25 16/in^2)	0 leakage
Air permeability	(maximum) 0.0
Abrasion (minimum cycle)	
Bar No. 15, 5 lb flexing bar load, 2.5 lb	(minimum warp) 50
Pressure plate load	(minimum filling) 50

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproval by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall time a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirement of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When specified (see 6.2), first article inspection shall consist of the examination and tests specified in 4.5. Samples shall be visually inspected for appearance, color, and finish.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated (see 6.2).

4.4.1 Examination of end item. Defects found during the examination shall be classified as specified in 4.4.1.1 through 4.4.1.5.

4.4.1.1 Yard-by-yard examination. Each roll in the sample shall be examined on face side only. When the total yardage in the roll does not exceed 100 yards (91.4 meters) the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards (91.4 meters) only 100 yards (91.4 meters) shall be examined. Defects defined in accordance with FED-STD-4, which are clearly noticeable at a normal inspection distance of 3 feet (0.91 meter) shall be scored and assigned demerit points as listed in 4.4.1.2 except that only the slubs and knots which exceed the limits shown on Sears Fabric Defect Scale or equal (see 6.6), "D" for slubs and "C" for knots shall be scored. No linear yard (increments of 1 yard (0.91 meter) on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than 4 points. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (rolls)[1]
3,200 or less	8
3,201 to 10,000	13
10,001 and over	20

[1] no more than one roll shall be taken from any shipping container unless the number of shipping containers in the lot is less than the required number of rolls in which case all shipping containers shall be present in the sample

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The lot shall be unacceptable if the points per 100 square yards (83.6 square meters) of the total yardage examined exceeds 50 points. The lot shall be unacceptable if the individual points per 100 square yard (83.6 square meters) of two or more individual rolls exceeds 75 points. If one roll exceeds 45 points per 100 square yards (83.6 square meters), a second sample, the same size as the final sample, shall be examined only for individual roll quality examination. The lot shall be unacceptable if one or more rolls in the second sample exceeds 45 points per 100 square yards (83.6 square meters). Point computation for lot quality and individual roll quality shall be as follows: l

$$\frac{\text{Total points scored in sample x 3600}}{\text{Contracted width of cloth (inches x total yards inspected}} = \text{Points per 100 square yards (83.6 square meters)}$$

4.4.1.2 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches (7.6 cm) or less in any dimension	= one point
For defects exceeding 3 inches (7.6 cm) but not exceeding 6 inches (15.2 cm) in any dimension	= two points
For defects exceeding 6 inches (15.2 cm), but not exceeding 9 inches (22.0 cm) in any dimension	= three points
For defects exceeding 9 inches (22.0 cm) in any dimension	= four points

The following defects, when present in the yard-by-yard and overall examination, shall be scored four points for each yard (0.9 meter) in which they occur:

a. Baggy, ridgy, or wavy cloth
b. Width less than specified
c. Uneven weaving
d. Objectionable odor
e. Overall uncleanness or discoloration
f. character of finish not equal to standard sample of first article when applicable.

4.4.1.3 Examination for length of individual rolls. The individual roll shall be exam.ined for gross length and the number of pieces in the roll. Any gross length found to be more than 2 yards below the gross length marked on the ticket or any roll found to contain more than one piece, when applicable, shall be considered a defect with respect to length. The unit of product for this examination shall be one roll. The sample unit for this examination shall be one roll (see 4.4.1).

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4.4.1.4 Examination for total yardage in sample. The lot shall be unacceptable if the total of the actual gross lengths of rolls in the sample is less than the total of the gross lengths marked on the roll tickets (see 4.4.1).

4.4.1.5 Examination for compliance with the Textile Fiber Products Identification Act. During the yard-by-yard examination, each piece in the sample shall be examined for compliance with the rules and regulations under the Textile Fiber Products identification Act. Each piece not labeled in accordance with this act shall be a defect. The lot shall be unacceptable if two or more of these defects occur (see 4.4.1).

4.5 Testing of the end item. The methods of testing as specified in accordance with FED-STD-191, whenever applicable and as listed in table II shall be followed. The physical values specified apply to the average of the determinations made on a unit of product from test purposes as specified in the applicable test method. The unit of product for test purposes shall be 2 continuous linear yards, full width of the finished cloth. The sample size shall be as shown in the list below. The lot size shall be expressed in units of 1 linear yard. The lot shall be unacceptable if one or more units of product fail to meet any test requirements specified.

Lot size (yards)	Sample size
800 or less	2
801 to 22,000	3
22,001 and up	5

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TABLE II. *Test methods.*

Characteristic		Requirement reference	Test method
Identification of fiber		3.21	[1,2]
Type of yarns		3.22	Visual
Fiber content		3.23	2510[1] ASTM D 578[1]
Weave		3.3	Visual
Colorfastness to:	Light	3.4.1	5660[3,4]
	Laundering	3.4.1	5610[4]
Crocking		3.4.1	5651[4]
Abrasion/resistance:	warp	3.10	5300[2]
	Filling	3.10	5300
Shrinkage		3.6	5550
Weight per square yard		3.10	5041
Yarns/inch		3.10	5050
Breaking strength:	Warp	3.10	5100
	Filling	3.10	5100
Tearing strength:	Warp	3.10	5132
	Filling	3.10	5132
Flame resistance (warp and tiling)		3.10	5903[4]
Hydrostatic (high)		3.10	5512
Air permeability		3.10	5450

[1] (See 6.5.1)
[2] End point is the state of destruction.
[3] The test shall be terminated if at the end of exposure of 20.0 standard fading hours, the material shows no appreciable change in color, or shall be compared with the standard sample, if applicable.
[4] Results should be shown as pass or fail.

5. PACKAGING

(The packaging requirements are not applicable to this specification)

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The woven cloth materials detailed in this specification are intended for use as portable two-panel smoke curtains and one-panel blankets.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2)
c. Type required (see 1.2)
d. Whether a first article sample is required (see 3.1 and 4.3)
e. Whether woven cloth is other than opaque (see 3.3)
f. Color required (see 3.4)
g. Width required (see 3.7)
h. Designs and patterns (see 3.8)
i. Whether length shall be other than 50 plus or minus 2-1/2 yards (see 3.9)
j. Quality conformance inspection (see 4.4)

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.1	DI-E-2121	Certification of Compliance	

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items should be a preproduction sample, a first article sample, a first production item, a sample selected from the first (specified number) of production items, a standard production item from the contractor's current inventory (see 3.2), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish

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evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Certification of compliance. Where certificates of compliance for fiber identification (see 3.2.1), finish (see 3.5), and fiber content (see 3.2.3) are submitted, the Government reserves the right to check test items to determine the validity of the certification (see 6.3).

6.5.1 Unless otherwise specified a certificate of compliance is acceptable for this requirement and should be submitted with each lot. (see 6.3)

6.6 Equal item. Prior to the use of an “or equal” item, the contractor shall submit the item with supporting data to the contracting officer for subsequent approval or disapproval by the responsible military agency.

6.7 Subject term (key words) listing.

Smoke barriers
Smoke containment

Preparing activity
Navy - SH
(Project 4210-N417)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-24757(SH)	9 November 1990

3. DOCUMENT TITLE

Cloth Fiberglass Coated

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
ıval Sea Systems Command (SEA 55Z3)	(1) Commercial (703) 602-7217	(2) AUTOVON (AV) 332-7217
c. ADDRESS (Include Zip Code) Washington, DC 20362-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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